FULLY FAITHFUL EXACT FUNCTORS REFLECT ISOMORPHISMS

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Uniqueness of cone triangle isomorphisms. lemma-cone-triangle-unique-isomorphism (same chapter, cited in-span; not carded). Gives the unique isomorphism of cone triangles closing the lifting argument.

Source reference. Original lemma 014B: sources/stacks-derived/pinned-original-source.tex, source lines 758–758. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $F : \mathcal{D} \to \mathcal{D}'$ be a fully faithful exact functor of pre-triangulated categories. Then a triangle (X, Y, Z, f, g, h) of $\mathcal{D}$ is distinguished if and only if $(F(X), F(Y), F(Z), F(f), F(g), F(h))$ is distinguished in $\mathcal{D}'$.*

Proof. The "only if" part is clear. Assume $(F(X), F(Y), F(Z))$ is distinguished in $\mathcal{D}'$. Pick a distinguished triangle (X, Y, Z', f, g', h') in $\mathcal{D}$. By Lemma 014B there exists an isomorphism of triangles

$$(1, 1, c') : (F(X), F(Y), F(Z)) \longrightarrow (F(X), F(Y), F(Z')).$$

Since F is fully faithful, there exists a morphism $c : Z \to Z'$ such that $F(c) = c'$. Then $(1, 1, c)$ is an isomorphism between (X, Y, Z) and (X, Y, Z'). Hence (X, Y, Z) is distinguished by TR1. □

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.